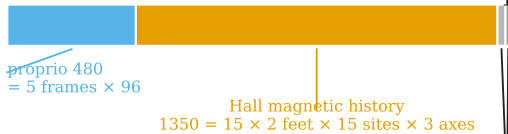
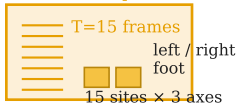


(a)

Observation $\mathbf{o}_t \in \mathbb{R}^{1864}$ 

Hall block expanded

proprio block expanded
96 per frame:

privileged (smr only): contact force,
ground μ , slip — critic & gate labels,
never in $\mathbf{o}_t$

(b)

Fast base + Hall-gated bounded residual

